# Conditional generation of visible language

## a Training

60 visible glyph images
Frozen causal reader

→

Continuous visual history
384 state + 1024 image field

History conditions every residual scale

Four noisy future cells
Independent or shared noise levels

→

Conditional strip denoiser
FiLM and cell attention

→

Velocity and visual losses
5.90M trainable parameters

## b Inference

Noise and visible history

→

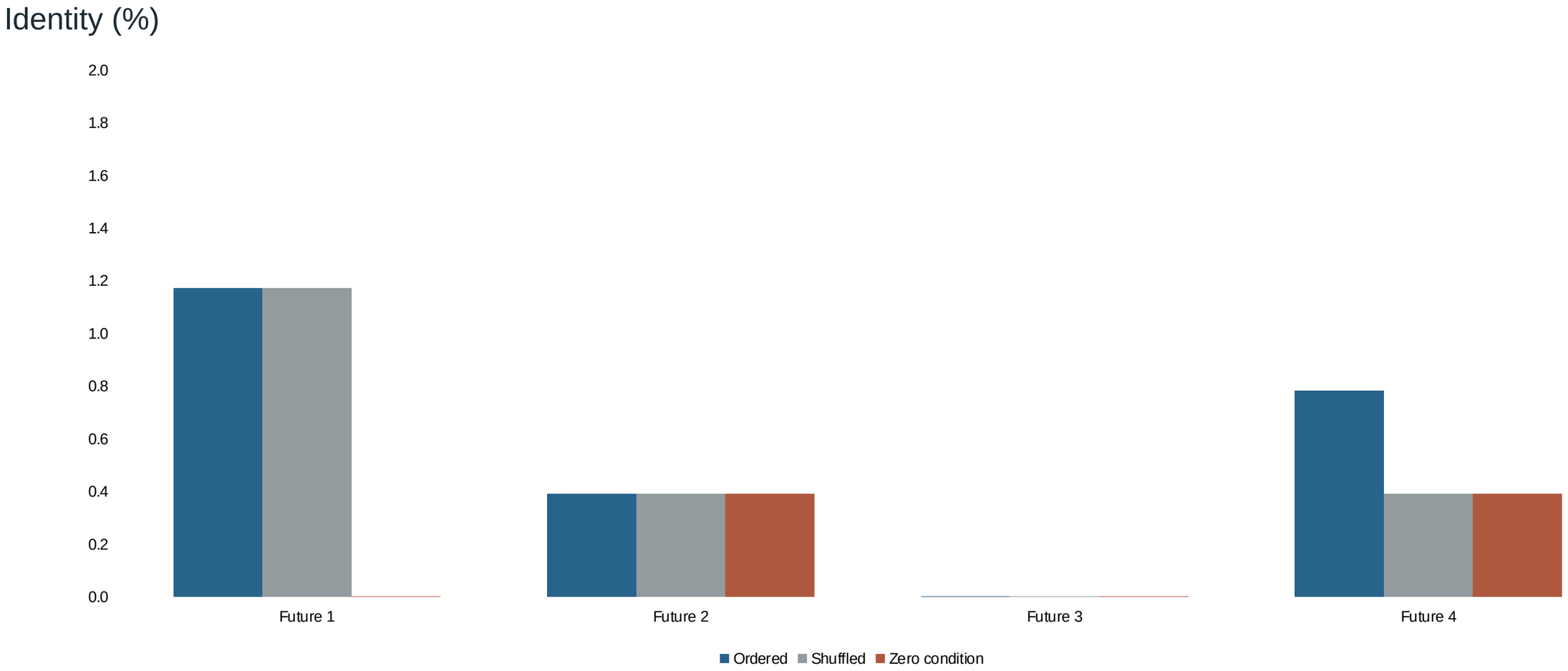
32-level reverse process

→

Four output images
Append pixels to next context

Source: V53 frozen protocol, implementation and locked audit. Schematic objects are editable.

# Direct block generation under paired visual histories




256 prefixes, one primary sample each, four output cells
Ordered block identity 0.586%, shuffled 0.488%, zero condition 0.195%

Source: locked V53 development report. Native chart includes its editable evidence workbook.

# Unselected output images expose the language error




Fixed first eight audit rows. Pixels are model output, with nearest-neighbour enlargement. No glyph replacement.

# Exact raster likelihood preserves a visible copy path

64 writing images
Frozen 16.28M reader

→

Continuous visual state
Hidden384 + image field1024

→

Fixed inverse field
Aligned signed pixel plan

## Training: exact Bernoulli image NLL

Earlier target rows
Four-layer row transformer

→

Earlier pixels within row
Strictly masked MADE

→

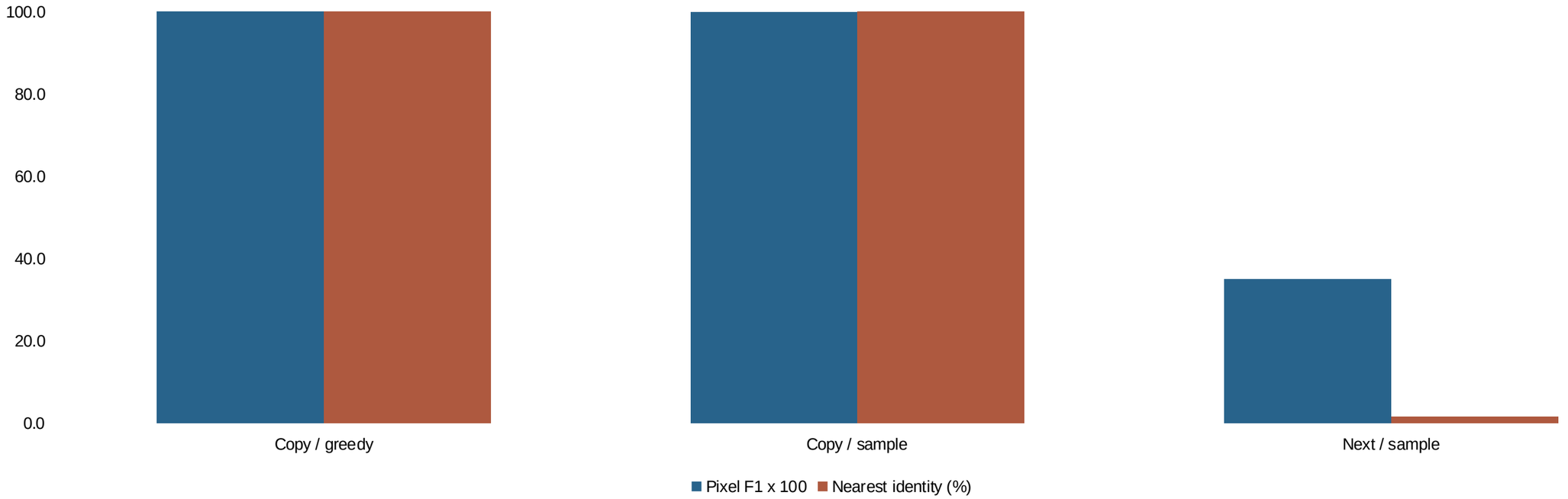
Aligned logit + residual
Normalized next-image law

25% supplied-image motor exercises; 75% natural predicted conditions. No glyph inventory.

## Inference: sample 1,024 causal pixels; append the visible raster to history

2.43M new density parameters; 18.71M total. Supplied-image copying is architectural, not language evidence.

# Copying succeeds; autonomous language remains unresolved




Exact-image copying: 256/256 greedy. Autonomous target identity: 4/256 ordered, 3/256 shuffled.

Target NLL: 15.219 ordered; 15.254 shuffled nats/image
Paired advantage 0.0355 nat; source-record bootstrap 95% CI [-0.1014, 0.1693]

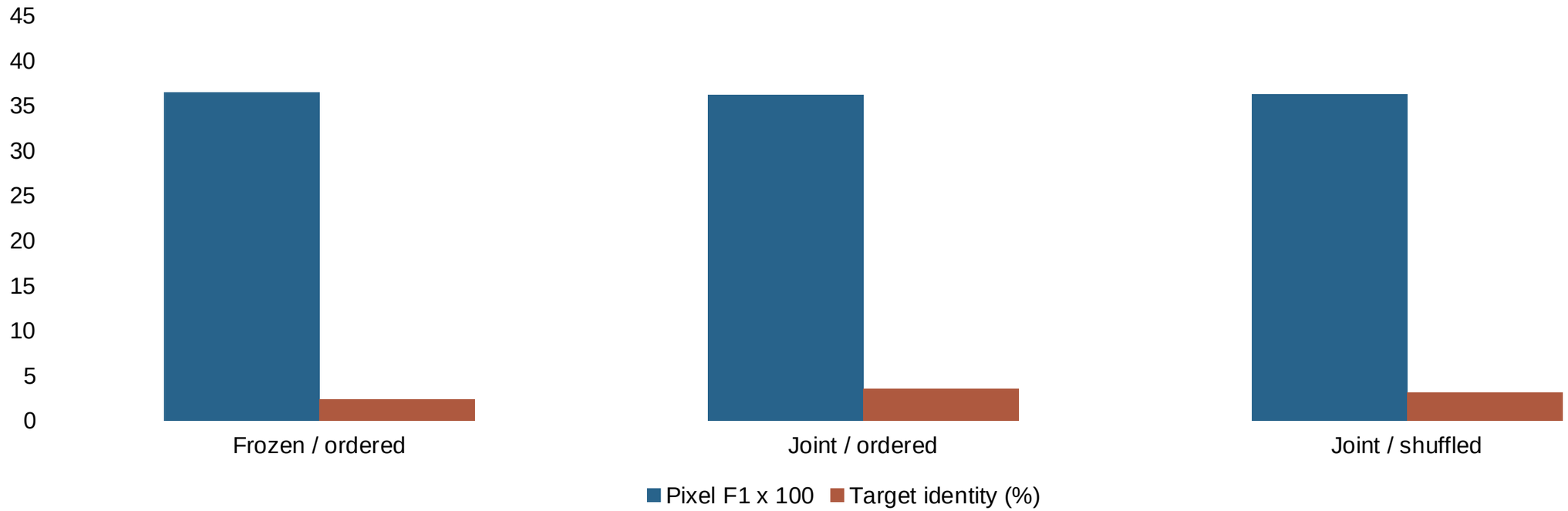
V54 locked development audit. Native chart and embedded workbook. Supplied-image controls reveal the answer.

# Actual rasters: exact motor copying and failed continuation




First eight fixed audit rows; nearest-neighbour enlargement. No OCR, image replacement or selected examples.

# Matched joint learning: better likelihood, incomplete generation

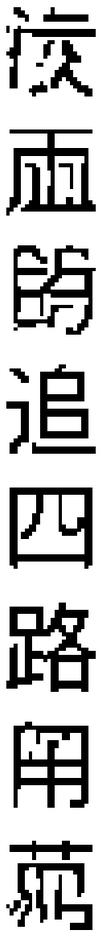
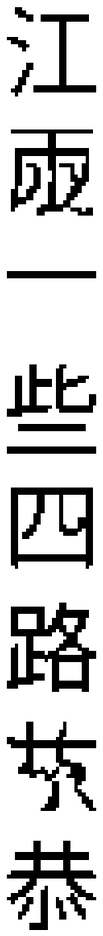
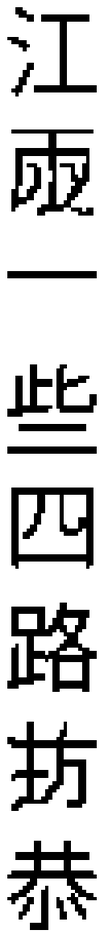



Exact target NLL: joint 7.026; matched frozen 7.579 nats/image
Paired gain 0.553; 95% source-record interval [0.450, 0.665]

Both arms retain exact supplied-image copying. Joint model passes 6/8 gates; generation remains unqualified.

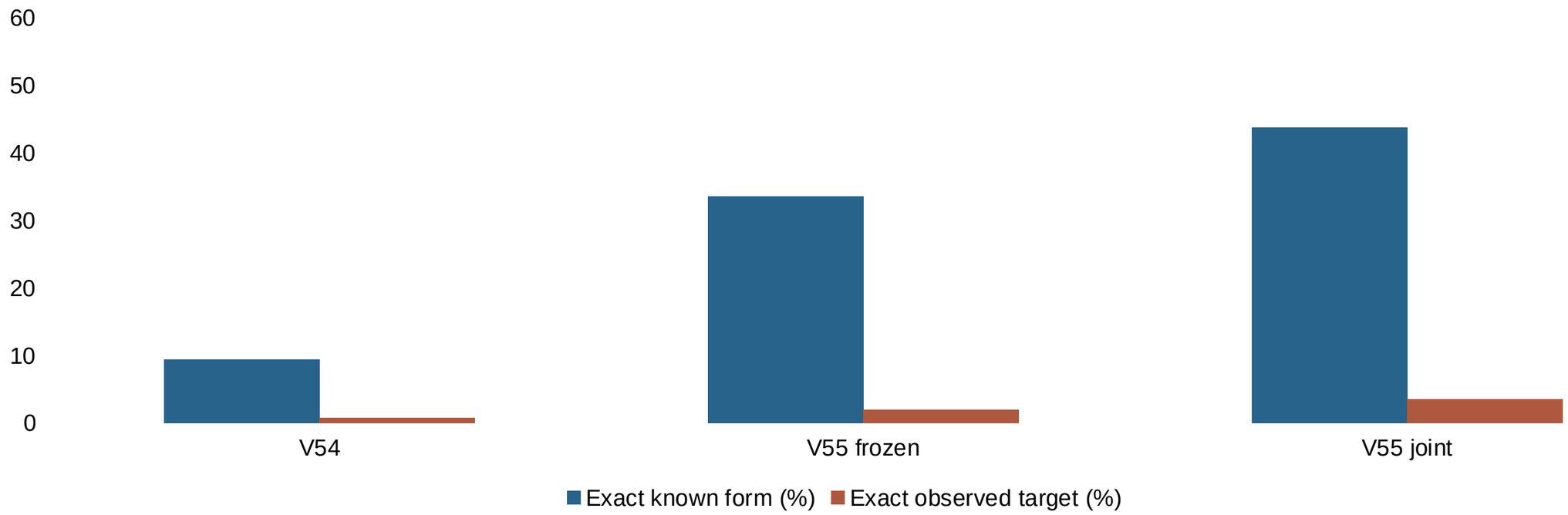
V55 locked matched audit; 512 NLL targets, 256 single-sample prefixes. Native chart includes its editable workbook.

# Raw matched output: exact copying and remaining language errors

| Target | Frozen sample | Joint sample | Joint shuffled | Joint greedy | Supplied copy |
|---|---|---|---|---|---|
|  |  |  |  |  |  |

First eight fixed audit rows, no selection. Pixels are actual output; no OCR, evaluator-bank replacement or external inference.

# Clear glyphs and correct language are separate tests

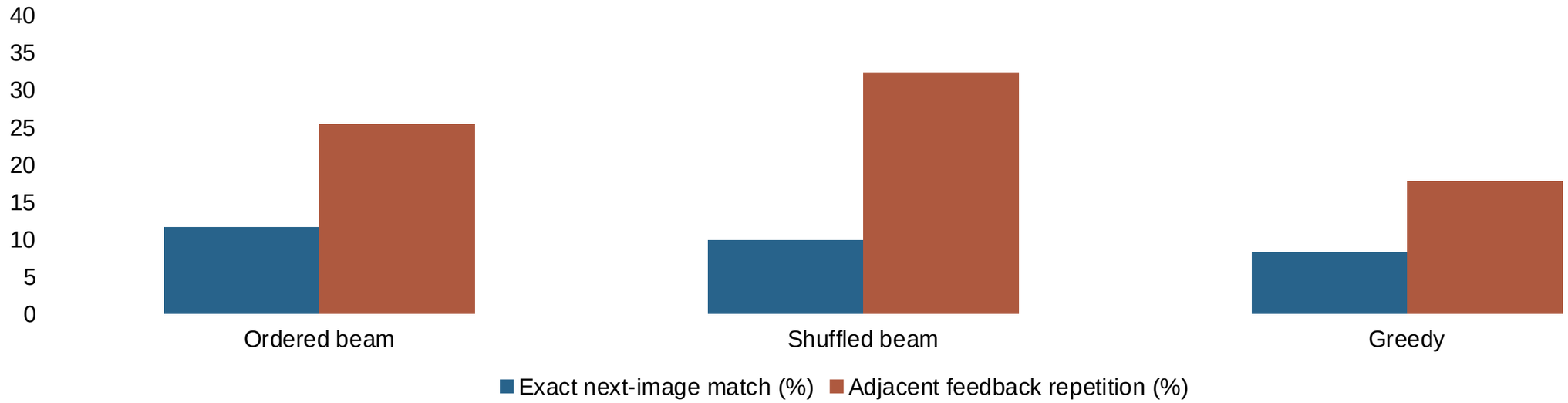



Exact known-form gain, joint minus frozen: 10.156 percentage points
95% source-record interval [4.219, 16.240]; 256 saved first samples

The fixed 9,932-image bank is used only for evaluation. No image is repaired or replaced.

Post-result diagnostic; all original qualification decisions remain unchanged. Native chart includes its editable workbook.

# Raw-pixel beam search: better immediate choice, repetitive output

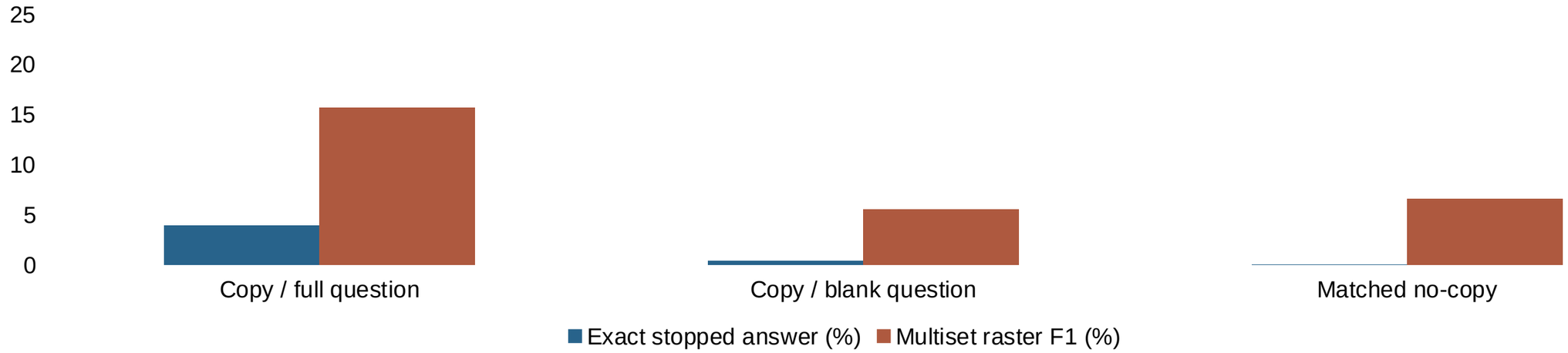



2,048 prefixes: beam gain over greedy 3.369 points [2.351, 4.412]; over shuffled 1.758 [0.966, 2.552]
Observed-stream repetition: 0.112%; beam: 25.446%. Rollout gate fails; decoder rejected.

| Observed future | Ordered beam | Shuffled beam | Original greedy |
|---|---|---|---|
| 不尬的就跑回来了 | 一个，不是一个， | 一个，不不不不不 | ，不不一，不不不 |
| 的女婿。丈人教我 | ，不是一人，不是 | ，不不不不不不不 | ，不不不不不不不 |

Unselected first two eight-step audit streams. Unchanged V55 weights; no vocabulary or repair. Both controls use highest FP32 precision.

# Natural Chinese questions: some binding, inaccurate complete answers




Question-dependent F1 gain: 10.165 points [7.043, 13.371]
Exact answers: 10/256; both questions correct: 1/393 pairs. Four of seven gates pass; model rejected.

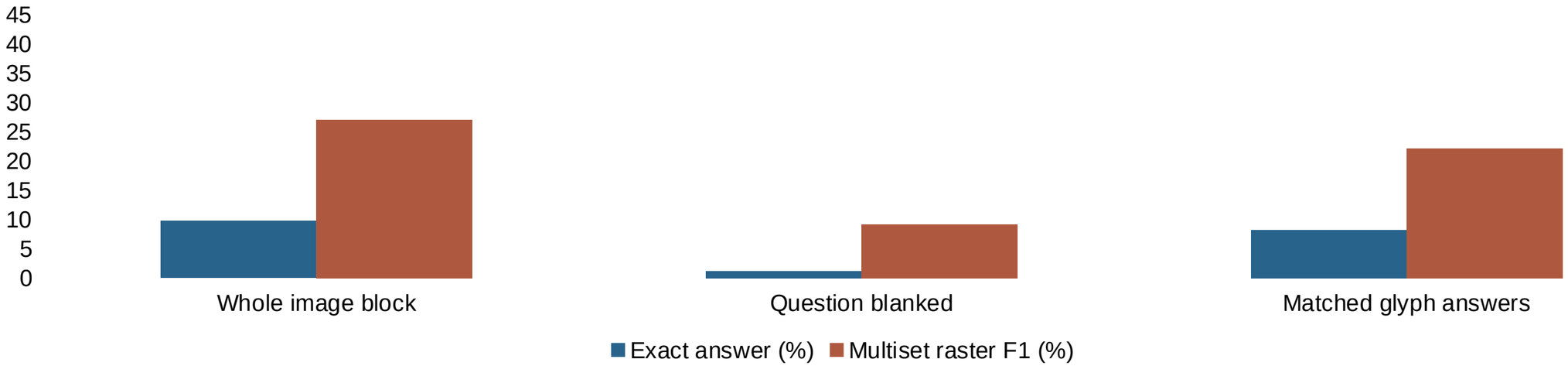
**Q1** １９４０年８月世宗大学更名成什么？

**Q2** 邵伯温是在什么时候去世？

| Reference 1 | Copy | Blank question | No-copy |
|---|---|---|---|
| 首尔女子师范学院<br>高宗绍兴四年（１１３４年） | １９４０年５月，Ｙ月世宗大学”<br>邵溥、邵博、邵传后》 | 节居绍圣、利州路转运作） | １９３０年<br>１３３０年 |

First two fixed audit rows; actual output images. White means zero emitted cells. All primary copy-arm answer cells use observed prompt images.

# Whole observed image answers: better F1, unresolved exact-match gain



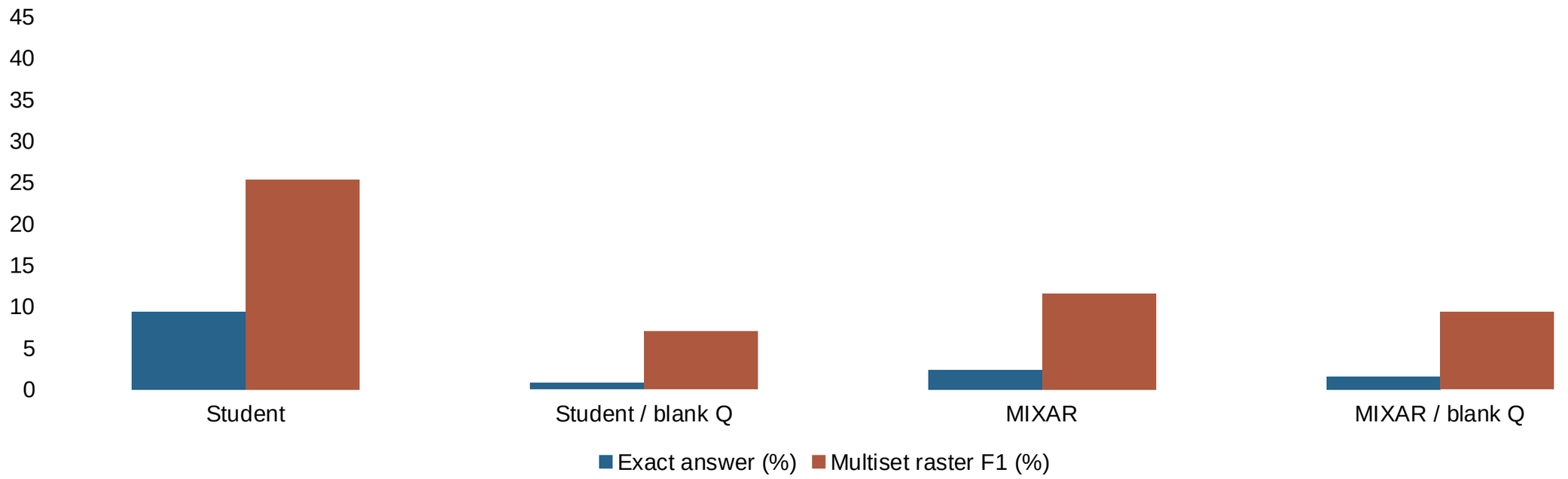

25/256 exact answers; matched glyph control 21/256. F1 gain 4.893 points [0.469, 9.047].
Only 4/393 paragraph pairs have both answers correct. Six of seven gates pass; reading rejected.

Prompt pixels → Causal reader → Start / end law → Source image answer

| | Retained reference | Actual block answer |
|---|---|---|
| Audit row 1 | 首尔女子师范学院 | “世宗学院” |
| Audit row 2 | 高宗绍兴四年（1134年） | 卒于高宗绍兴四年（1134年） |

First two fixed audit rows. Copying preserves selected pixels; it does not certify the meaning of the selected text.

# Frozen visual foundation probe: larger pretrained features do not help here

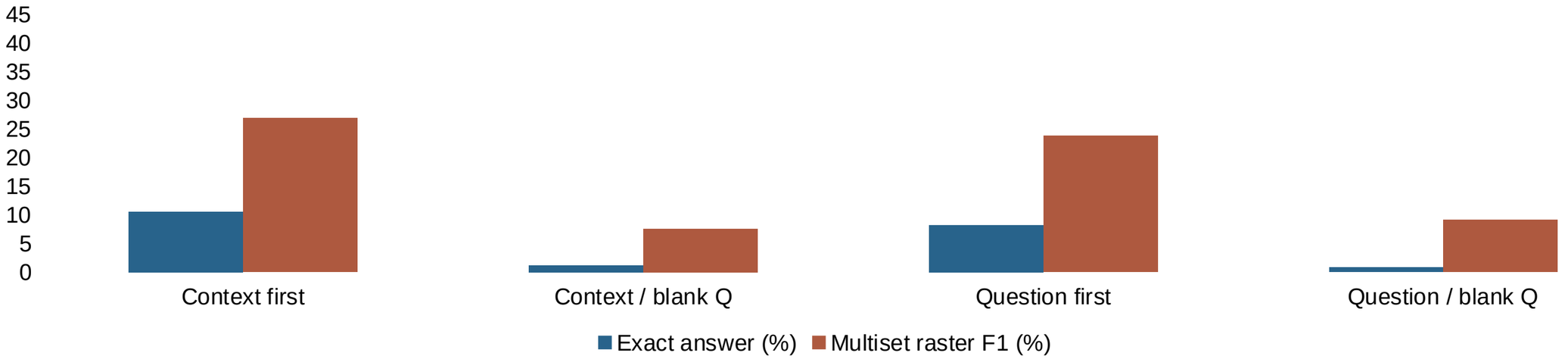



Identical 789,520-parameter heads; 4,000 matched updates each.
Student 24/256 exact; MIXAR 6/256. MIXAR-minus-student F1 −13.752 points [−18.247, −9.094].

MIXAR has different pretraining and native rendering; this probe uses our Noto CJK grid.
External four-H200 pretraining is not one-4090 work. Frozen readout comparison rejected; not a general encoder ranking.

Continuous prompt-image features only. Shared cached release weights; no external inference call or answer annotation enters feature preparation.

# Does reading the question first improve evidence selection?




**Context first**

Passage images X → Question images Q → Answer

**Question first**

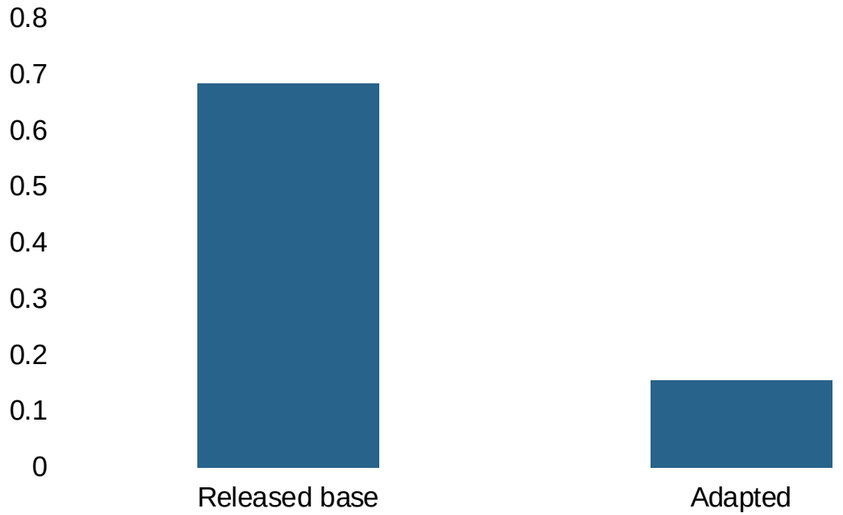
Question images Q → Passage images X → Answer

Question-first minus context-first F1: -3.143 points [-6.544, 0.446].
4/8 registered gates pass; decision: reject_question_first_intervention.

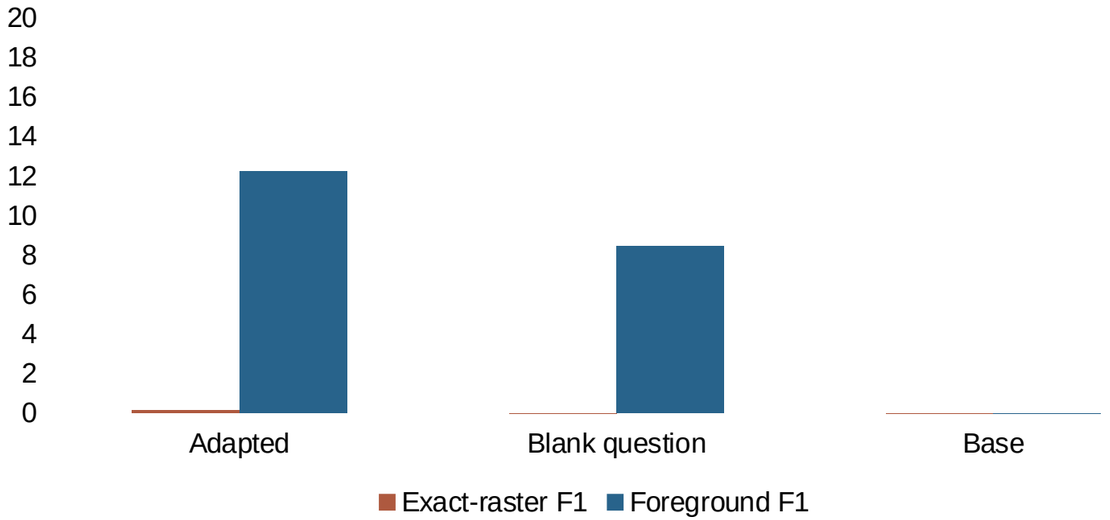
Same checkpoint, data, parameters and training budget. Source pixels only; reused development, not a sealed final test or general language qualification.

# Native free writing: improved pixel fit, failed language generation

## Held-out answer-pixel NLL (nats)




## Generated image scores (%)



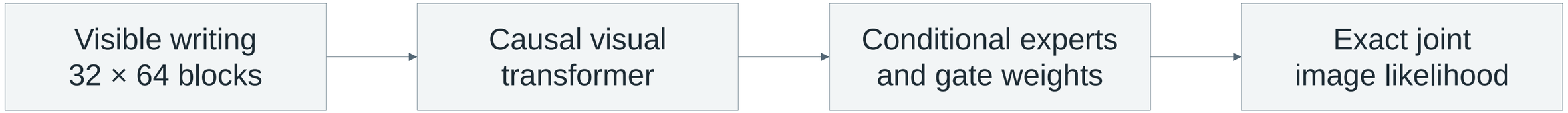

0/256 exact answers and 0/393 pairs both correct. Five of ten registered gates pass.
477.15M externally pretrained parameters, fully adapted for 1,000 updates on GPU 0.

| | Retained reference | Actual free-pixel answer |
|---|---|---|
| Audit row 1 | 首尔女子师范学院 | |
| Audit row 2 | 高宗绍兴四年（１１３４年） | |

Training: 1,501.1 s and 9.912 GiB. Offline native-PNG replay passes. External pretraining and reused development disclosed.

# Conditional whole-image density

## Training: prompt and observed answer images




$$p(y \mid h) = \Sigma k\ \pi k(h)\ \Pi i\ \text{Bernoulli}(yi\ ;\ \text{sigmoid}(aki(h)))$$

One shared expert choice couples all pixels in a bitmap. Each expert is a neural function of context.
K = 1: 477.15M parameters. K = 8: 490.00M. Same examples and update budget; different head size.

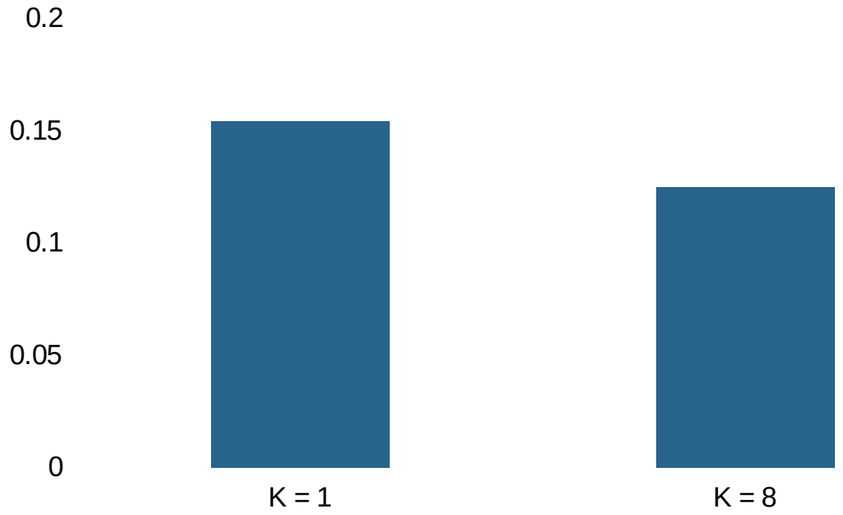
## Inference: image prompt → candidate bitmaps → score under full density → physical pixels

Feed generated pixels back into the visual context; stop at an all-white image block.
Component-mode search is approximate MAP. No character bank, OCR, Unicode IDs or external LM in the actor.

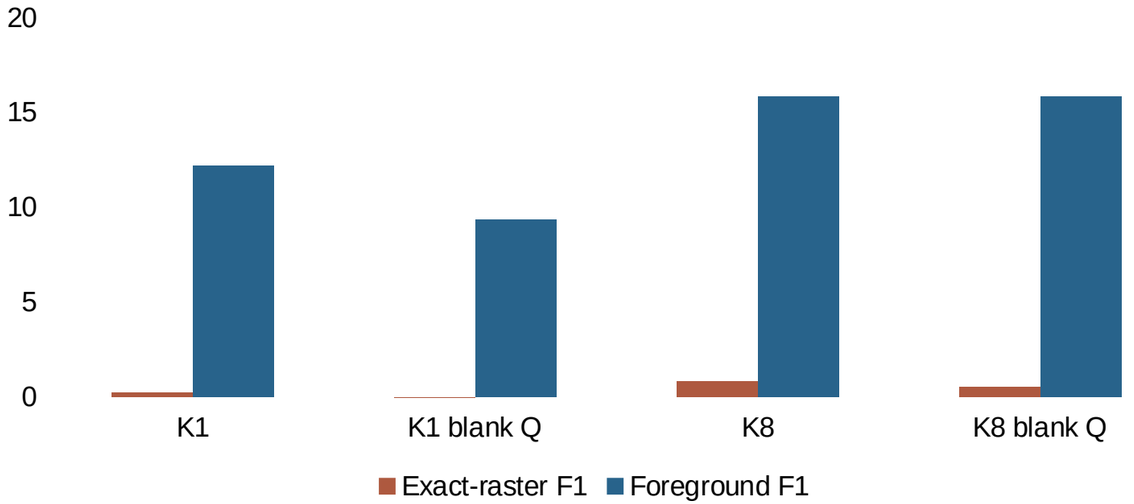
Externally pretrained MIXAR foundation. The mixture law is a tested hypothesis, not a guarantee of readable language.

# Conditional density: held-out fit and generated writing

## Held-out answer-pixel NLL (nats)




## Generated image scores (%)




Exact answers K1 / K8: 0 / 0 of 256. Both-correct pairs: 0 / 0 of 393.
Registered decision: reject density intervention. General image-language quality remains unqualified.

| | Retained reference | K = 1 | K = 8 |
|---|---|---|---|
| Audit row 1 | 首尔女子师范学院 | [illegible] | 1 9 4 0 年 |
| Audit row 2 | 高宗绍兴四年（1 1 3 4 年） | [illegible] | 1 9 [illegible] |

First two fixed primary answers, unrepaired; white display padding. One seed, reused development, external pretraining disclosed.

# Question-conditioned image-language training

## a Observed training images: two questions about one paragraph

**Question region Q0**

哪个人是世纪越南反唐起义军的军事首领？

Recorded answer Y0 冯兴

**Question region Q1**

冯兴与他的弟弟冯骇接受谁的建议，率众起义？

Recorded answer Y1 杜英翰

Each full prompt Xi also contains the same paragraph image. Appended white termination blocks are not drawn.

## b Four causal likelihoods under the same parameters

(X0, Y0) (X0, Y1)
(X1, Y0) (X1, Y1)

→

Shared image generator
490M parameters, 8 experts

→

| | Y0 | Y1 |
|---|---|---|
| X0 | $e_{00}$ | $e_{01}$ |
| X1 | $e_{10}$ | $e_{11}$ |

eij = mean negative log likelihood of the complete answer suffix, including termination

## c Positive image likelihood with paired question assignment

$$D = e_{01} + e_{10} - e_{00} - e_{11}$$

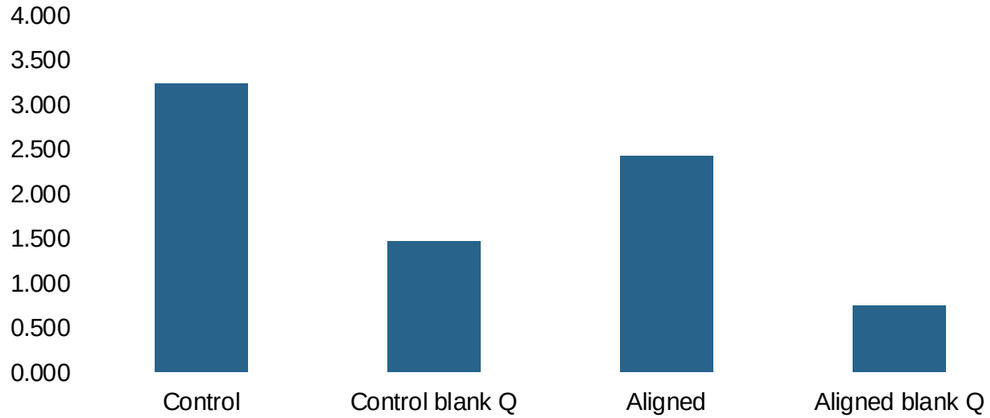
λ = 0: control; λ = 1: assignment

$$L = (e_{00} + e_{11}) / 2 + \lambda \cdot 0.01 \cdot \mathrm{softplus}(-D / 0.01)$$

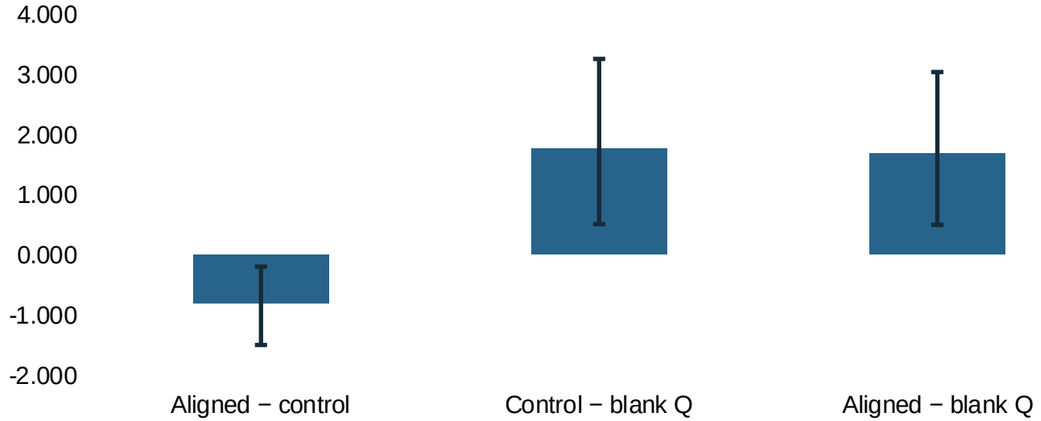
Observed targets, not model outputs. Both arms start from the same V62 EMA; autonomous answers are evaluated separately.

# Question assignment: completed free-writing test

## Strict-raster F1 (%)




## Paired differences with 95% intervals (pp)




**Exact answers: control 2/256, aligned 1/256. Both-correct pairs: 0/393 in each arm.**

### First two fixed primary answers

1: Reference 首尔女子师范学院

1: Control



1: Aligned



2: Reference 高宗绍兴四年（1134年）

2: Control



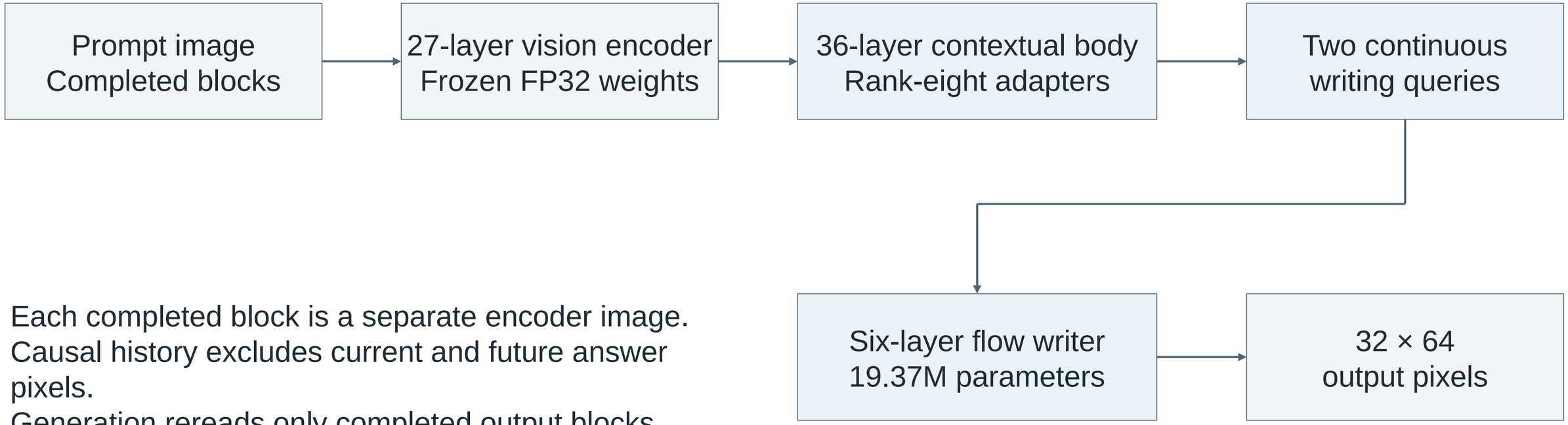
2: Aligned



Unrepaired pixels and white padding. One seed, reused development and external pretraining. Both reading gates fail.

# Complete continuous image actor

7.548B total parameters  25.58M trainable  17.181 GiB peak allocated




Each completed block is a separate encoder image.
Causal history excludes current and future answer pixels.
Generation rereads only completed output blocks.

$$Yt = (1 - t)\,U + t\,Y \qquad U \sim N(0, I) \qquad Lflow = E\,||v(Yt,t,h) - (Y - U)||^2$$

Frozen contextual weights use BF16 storage with FP32 computation. A separate classifier predicts stopping.

**Execution checks pass. Writing and language quality remain unqualified.**